it allows the OS to keep frequently used parts of the running program in memory (as well as parts of the OS itself).

It is always better to assign a dedicated partition on a hard disk to the virtual memory. Since hard disks are much slower than RAM, it takes a significant amount of time to move data to and from virtual memory. Having a separate partition causes reading and writing of virtual memory to be independent of the reading and writing of other files. We will see how this scheme can be improved further.

## Improving performance through cross swapping

If you have more than one hard disk, you can improve the performance of the virtual memory further. To do so, simply install the OS on one hard disk and the virtual memory on a dedicated partition on another hard disk. This scheme is much better than having both on the same hard disk because it gives independent control to the virtual memory. While files are being read from or written to one hard disk, the OS can operate the virtual memory simultaneously on the other—it won't have to wait for either operation to complete before starting the other.

These days, it is becoming increasingly common to have more than one OS installed on a computer and use (boot) any one of them at a time. If you have more than one hard disk, you can use cross swapping to utilise all hard disks optimally.

It is best to install separate OSs on separate hard disks. This isolates their partition tables, provides physical encapsulation, and prevents space conflict and re-installation if you want to change partition sizes later. However, typically, only one OS is used at a time. All the others remain unused as long as that one is being used. If the other systems are on different hard disks, those hard disks will remain unused for that time. We can therefore utilise them for the virtual memory of the OS that is running. To do this, simply allot each OS's virtual memory to a partition on a hard disk other than that on which the OS is installed. We will look at a scenario where Microsoft Windows and Linux are to be dual-booted.

Let's suppose we have two hard disks, A and B. We install Windows on hard disk A and a distribution of Linux on hard disk B. Now, to employ cross swapping, we place the virtual memory for Windows on a dedicated partition on hard disk B, while we place the virtual memory for Linux on a dedicated partition on hard disk A. However, to achieve this, it is best to create the partitions before or while installing the operating systems. They can be created later, but that requires either a RAID/LVM set-up or special software. If you wish to follow the process mentioned, set up your hard disk partitions as shown in Figures 1 and 2.

Figure 1 shows hard disk A with the first two partitions allotted for Windows files and the last partition used as dedicated virtual memory for Linux (the swap partition). Figure 2 shows a similar arrangement of hard disk B with a virtual memory partition for Windows and the last two partitions for Linux files. The exact positioning of the

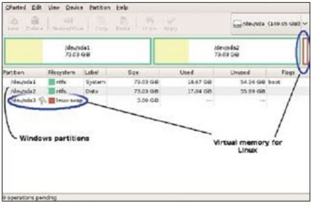


Figure 1: Hard disk A with Windows partitions and a virtual memory partition for Linux

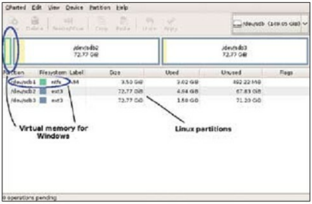


Figure 2: Hard disk B with Linux partitions and a virtual memory partition for Windows

partitions is such because Windows always needs the first partition of the first hard disk, and the Windows swap partition is placed at the head of the second disk because Windows sometimes achieves better performance if its virtual memory partition is placed at the beginning of a hard disk. The two partitions each for Windows and Linux files are just my personal preferences to keep the system and personal files separate. You can use one or more partitions in their place for each OS.

To actually create such a layout during installation, start with two blank hard disks and follow the steps below (note: this assumes we're using Windows XP and so the exact steps may differ for other versions—refer to Windows documentation for changing the amount of virtual memory in such cases).

1. Create a primary partition on the first hard disk, make it bootable (active), and install Windows on it (you can create the partition from within the Windows installation program itself).
2. If you want another Windows partition for your personal files, as shown in Figure 1, create a primary partition on the first hard disk (Disk 0) using the Windows disk management tool (right click on *My Computer*, click *Manage→Disk Management*).
3. Leave the area for the Linux swap unpartitioned at the end of the first hard disk (Disk 0). Keep about 1.5 times the amount of your RAM reserved for this space. You